Do Now · retrieval
LMC instructions and data

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does direct addressing [30] retrieve?

Q2

What is the accumulator?

Q3

What is a mnemonic?

Q4

Search for 19 in sorted [1,4,7,10,13,16,19,22,25]. Show midpoint indexes checked.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Know LMC's parts — LMC uses numbered mailboxes to hold instructions and data, and an accumulator for working values.

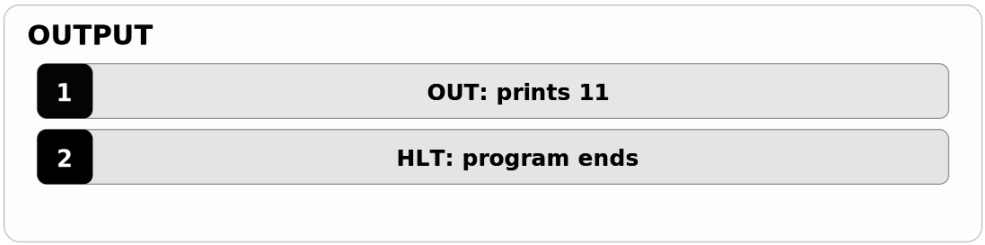
Core idea 2

Trace instructions in order — First INP enters 4 into ACC, then STA FIRST copies 4 to the data mailbox before another input overwrites ACC.

Core idea 3

Produce the result — Second INP makes ACC=7; ADD FIRST adds stored 4 to make 11, and OUT prints 11. HLT ends the program.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LMC — Little Man Computer teaching instruction set.

ACC — Accumulator holding current working value.

mailbox — Numbered LMC memory location.

STA — Store accumulator to memory.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

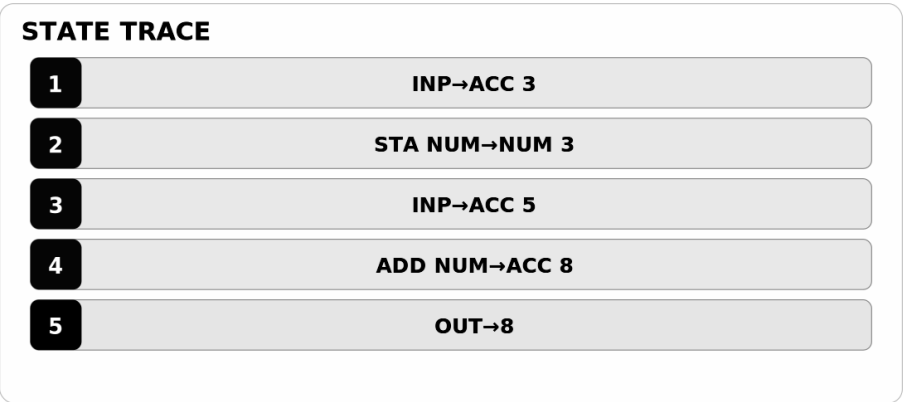
Problem

Write and trace an LMC program adding inputs 3 and 5.

Follow the worked method

1. Predict — Store the first input 3 into NUM so the second input will not overwrite it.
2. Explain — Read the second input 5, add NUM to get 8 and output the result.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Store input 9 as A and input 4 as B, since the accumulator is overwritten by INP.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain what INP, OUT, LDA and STA do in LMC.

Question 2

Trace INP; STA X; INP; ADD X; OUT; HLT; X DAT 0 using inputs 6 and 8. State each change and output.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write an OCR LMC program that takes two inputs from 0 to 999, with the first greater than or equal to the second. Subtract the second from the first and output the non-negative result. Trace inputs 9 and 2.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In LMC, STA X replaces ACC with the stored value of X.

Use key vocabulary from memory

1. Explain LMC in your own words.

2. Explain ACC in your own words.

3. Explain mailbox in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.